

Screw Versus Suture Bridge Fixation for Medial Femoral Condyle Osteochondritis Dissecans Demonstrate Comparable Healing Rates

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Background: Fixation of unstable medial femoral condyle (MFC) osteochondritis dissecans (OCD) lesions aims to achieve osseous union while preserving the articular cartilage. Suture bridge fixation has emerged as an alternative to bioabsorbable implants or metallic headless compression screws.

Hypothesis: The authors hypothesized that rates of union, conversion to a secondary cartilage restoration procedure, and patient-reported outcomes would not differ significantly between screw and suture bridge fixation, with fewer implant-related complications after suture bridge fixation.

Study Design: Cohort study; Level of evidence, 3.

Methods: A cohort of consecutive patients at a single center with unstable MFC OCD lesions treated with metallic headless compression screws or suture bridge constructs was studied using prospectively collected data from the multicenter Research in Osteochondritis of the Knee (ROCK) registry. Inclusion required a minimum of 6-month follow-up. Patient and lesion characteristics and surgical variables were recorded. Radiographic and advanced imaging outcomes, including magnetic resonance imaging-based healing criteria, were assessed to determine union status. Reoperation, secondary articular surface changes, conversion to a secondary cartilage restoration procedure, and patient-reported outcomes were compared between groups.

Results: A total of 69 lesions were included, with a mean patient age of 14.3 ± 2.4 years. Final union status and conversion rates to a secondary cartilage restoration procedure were similar between screw (17.2%) and suture bridge (20%) fixation ($P \geq .999$), with no difference in timing of conversion (12.7 ± 8.7 vs 12.6 ± 9.2 months, respectively; $P = .833$). Implant removal rates were similar between screw (89.7%) and suture bridge (81.3%) groups ($P = .478$); however, screw fixation was associated with earlier time to permanent implant removal (5.1 ± 4.1 vs 8.5 ± 7.8 months; $P = .019$), and more severe secondary articular surface changes, including International Cartilage Repair Society grade 4 defects (10.7%) observed exclusively in the screw cohort ($P < .001$). No differences were observed between groups in improvement from baseline or final Pediatric International Knee Documentation Committee, Knee injury and Osteoarthritis Outcome Score subscores, or Pediatric Functional Activity Brief Scale scores.

Conclusion: Metallic headless compression screw and suture bridge fixation demonstrated comparable union rates, equivalent improvements in patient-reported outcomes, and no statistically significant difference in conversion to a secondary cartilage restoration procedure in unstable MFC OCD lesions. Suture bridge fixation allowed longer implant retention with reduced secondary articular surface damage.

Keywords: headless compression screws; medial femoral condyle; osteochondritis dissecans; patient-reported outcome measure; suture bridge

Osteochondritis dissecans (OCD) is a focal, idiopathic alteration of subchondral bone and/or its precursors with risk for instability and disruption of adjacent articular cartilage that may result in premature osteoarthritis.⁸ OCD is a relatively uncommon condition and most commonly presents in skeletally immature and adolescent patients, with reported incidence rates^{4,14,25} ranging from 9 to 29 per 100,000. Various zones of propensity have been described, with the lateral aspect of the medial femoral condyle (MFC) considered to be the most common.^{14,23,24,31}

Management of knee OCD lesions ranges from nonsurgical, conservative measures such as activity modification or weightbearing restriction, to surgical cartilage replacement techniques, and is guided by skeletal maturity, lesion characteristics, symptom severity, and response to previous treatments.⁴ Specifically, fixation of unstable OCD lesions aims to achieve ossification and union of the progeny fragment to the parent bone while preserving joint congruity and the health of the articular cartilage. Traditional fixation methods utilizing metal screws provide rigid compression. Still, they are associated with potential disadvantages, including articular surface violation, imaging artifact, implant loosening, and the possible indication for early hardware removal.²⁶ Bioabsorbable screws and chondral darts mitigate imaging and implant removal concerns. However, variable degradation profiles and cyst formation, potential loosening and secondary wear, inflammatory reactions, and unpredictable mechanical integrity are reported^{2,22,33,36} (Figure 1).

Suture bridge fixation has emerged as an alternative strategy for the management of osteochondral injuries, including OCD and acute osteochondral fractures across multiple anatomic locations, such as the knee, elbow, and shoulder.^{3,11,12,15,16,21,29,35} Proposed advantages include more uniform load distribution across softer lesions and those with lesser bone in the progeny, and a reduced risk of catastrophic fixation failure and possibly lesser risk of opposing articular surface wear. These characteristics may be particularly advantageous in complex cartilage repair scenarios involving displaced or chronic lesions with reduced progeny material integrity, or with all-chondral lesions.

The primary purpose of this study was to compare imaging-based union outcomes of unstable MFC OCD lesions treated with traditional metallic headless compression screws versus suture bridge fixation. A secondary purpose of this study was to compare differences in complications, including conversion to a secondary cartilage restoration procedure and the occurrence and severity of adjacent articular surface damage.

We hypothesized that rates of union, conversion to a secondary cartilage restoration procedure, and patient-reported outcome measures (PROMs) would not differ significantly between screw and suture bridge fixation techniques, with fewer implant-related complications after suture bridge fixation.

METHODS

Study Design and Patient Selection

A comparison of prospectively collected data from consecutive patients with unstable MFC OCD lesions treated with either screw or suture bridge fixation at a single center in the multicenter Research in Osteochondritis of the Knee (ROCK) registry was accomplished. All procedures were performed between November 2016 and December 2024, and a minimum of 6-month follow-up was required for inclusion in the study. Suture bridge fixation for OCD management was introduced into clinical practice at the study institution in 2019. Eligible patients included those with unstable MFC OCD lesions, identified either on preoperative imaging or at the time of arthroscopic evaluation, and treated at the surgeon's clinical discretion with fixation using either metallic headless compression screws or suture bridge techniques. Lesion instability was defined per previously published ROCK criteria, including the presence of ≥ 1 of the following on preoperative magnetic resonance imaging (MRI): a high-signal rim surrounding the lesion, a discrete chondral breach, a subchondral cyst >5 mm, or a physeal line crossing the lesion. Intraoperatively, instability was confirmed by arthroscopic probing demonstrating fragment mobility or frank displacement.^{5,6,10,25} Both unilateral and bilateral knees were included, with each knee treated as a separate analytic unit when applicable. Patient characteristics and procedural data were collected from the electronic medical record (Epic Systems). Radiographic and advanced imaging data, follow-up cadence, and PROMs were prospectively collected through the multicenter ROCK registry.

Radiographic and Advanced Imaging Assessment

Pre- and postoperative imaging were reviewed for all patients. Standard knee and low-dose full-length lower extremity standing radiographs (EOS Imaging) were used to assess the MFC OCD and coronal limb alignment, respectively. Preoperative MRI was reviewed to

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Ethical approval for this study was obtained from Scottish Rite for Children (STU 062017-100).

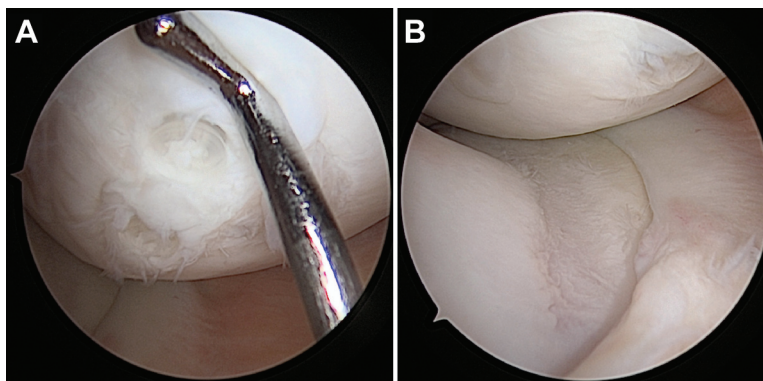


Figure 1. Arthroscopic image of bioabsorbable screw fixation of MFC OCD lesion with subsequent backout and secondary wear of tibial articular surface. (A) Bioabsorbable screws are noted on the MFC, which are prominent with surrounding chondromalacia. (B) Note the full-thickness chondral defect on the medial tibial plateau attributed to the bioabsorbable screws. MFC, medial femoral condyle; OCD, osteochondritis dissecans.

characterize all lesions, including surface area, depth, bone marrow edema, and qualitative features, including presence of subchondral cyst formation, the Oreo cookie sign (tri-laminar alternating low and high signal density suggesting fibrous interposition between parent and progeny bone), the omen sign (radially-oriented low signal within the articular cartilage contour that may represent early cartilage fissure and instability), bone within the progeny fragment, and surrounding bone marrow edema. Lesion surface area was calculated using the largest orthogonal measurements obtained on sagittal and coronal MRI sequences.¹⁰ Lesion depth was defined as the maximal distance from the articular surface to the deepest demarcation of the lesion on either sagittal or coronal imaging. Bone marrow edema was quantified as a percentage of the MFC involved, based on visual estimation in the coronal and sagittal plane cuts of the MRI demonstrating the mean major length and width of the OCD lesion. It was recorded as either <25%, 25%-50%, or >50% of the condyle. Final postoperative imaging modality, either MRI or computed tomography (CT) scan, was determined by the treating surgeon's clinical judgment.

Surgical Technique

All procedures were performed by 1 of 2 fellowship-trained orthopaedic sports medicine surgeons (H.B.E. and P.L.W.). Lesions were evaluated arthroscopically and treated according to lesion characteristics and surgeon discretion. Fixation was accomplished utilizing either a metallic headless compression screw or suture bridge construct with either absorbable (No. 1 or 2 Vicryl) or nonabsorbable suture (1.3 mm SutureTape; Arthrex) using a previously described technique.^{9,34,35} All patients with screw or nonabsorbable suture constructs were recommended for hardware removal after confirmation of union on postoperative advanced imaging. Implant removal was not indicated for lesions treated with absorbable suture.

Both arthroscopic-assisted and open approaches were utilized for screw and suture bridge fixation. Approach selection was determined by lesion size, location, degree of fragment displacement, and adequacy of arthroscopic visualization for reduction and fixation. Bone grafting was performed when debridement of the lesion bed resulted in insufficient osseous material in the parent to maintain reduction of the progeny fragment and restore a congruent articular contour. For suture bridge constructs, fixation was achieved using 2.9-mm PushLock anchors (Arthrex) with either absorbable or nonabsorbable suture. The choice between absorbable and nonabsorbable suture was guided by lesion type, severity of preoperative imaging findings, and degree of intraoperative instability. Standard lesion preparation included a combination of drilling, debridement of fibrous tissue, and/or bone grafting (cancellous autograft harvested from the proximal tibia, distal femur, and/or iliac crest based on required volume and surgeon judgment). Representative imaging and intraoperative photographs demonstrating MRI appearance and suture bridge fixation technique are shown in Figure 2.

After fixation, patients were made nonweightbearing for approximately 6 weeks while allowing range of motion. Patients then progressed to as-tolerated weightbearing with low-demand activities of daily living for an additional 6 weeks, with gradual activity advancement thereafter. Return to lower extremity impact sports was restricted for a minimum of 6 months postoperatively. Progression was individualized based on clinical symptoms and imaging evidence of union.

Outcome Measures

Primary outcomes included imaging-based union status on radiographic and advanced imaging modalities as well as conversion to a secondary cartilage restoration procedure. As part of clinical practice after OCD fixation, early

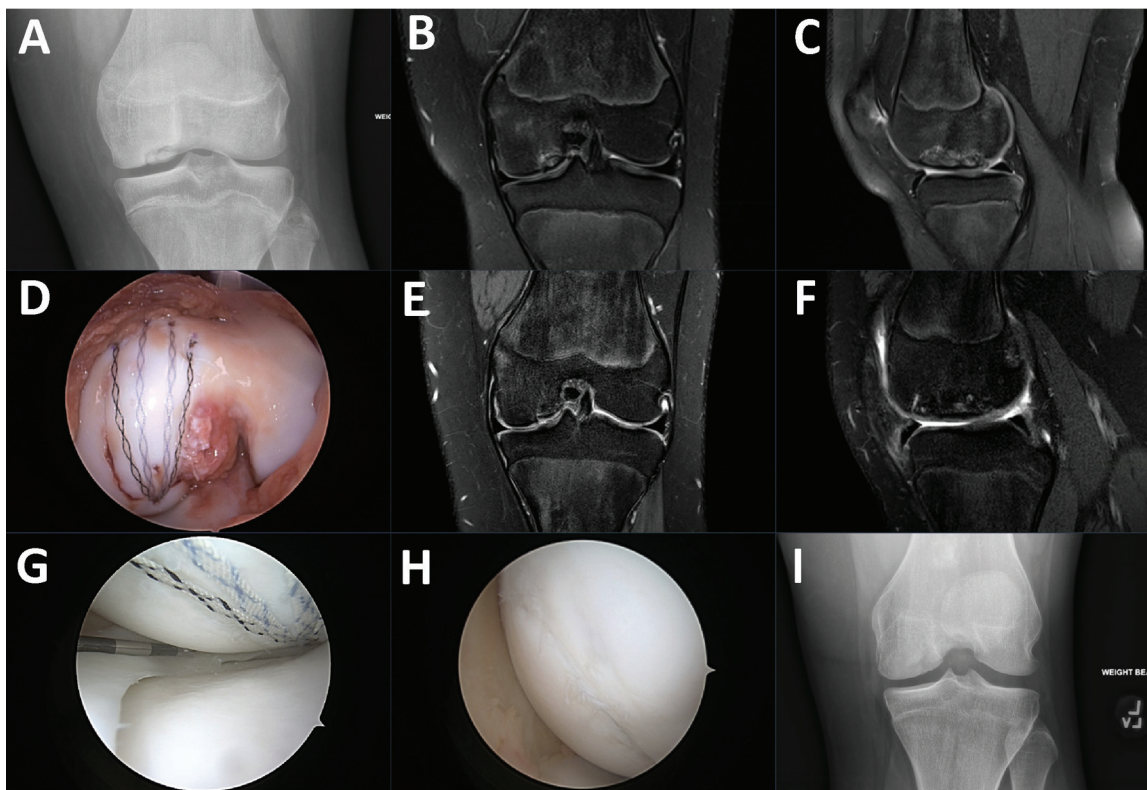


Figure 2. Progression of a left MFC OCD lesion from presentation to completion of union after open suture bridge fixation with nonabsorbable suture tape. (A) AP plain radiograph demonstrating MFC OCD lesion. (B and C) PD TSE FS MRI coronal and sagittal sequences, respectively, demonstrating MFC OCD lesion. (D) Intra-operative photograph of open suture bridge fixation technique. (E and F) 4.5 months postoperative PD TSE FS coronal and sagittal imaging sequences, respectively. (G) 5.5 months postoperative arthroscopic images showing previous MFC OCD with suture bridge fixation in place and clear view of adjacent tibial articular surface and (H) after removal of sutures. (I) Radiographic notch view of healed MFC OCD lesion 11 months after index operation. AP, anteroposterior; MFC, medial femoral condyle; MRI, magnetic resonance imaging; OCD, osteochondritis dissecans; PD TSE FS, proton density turbo spin-echo fat-suppressed.

postoperative advanced imaging was obtained around 3 to 6 months after surgery. A repeat advanced imaging study (MRI or CT) was obtained selectively at approximately 1 year postoperatively in patients who had not demonstrated satisfactory clinical or imaging progress toward union, based on persistent symptoms, lack of progressive union on the most recent plain radiograph, or advanced imaging study.

Union of the lesions was categorized as follows: full union, defined as 100% continuity of cancellous or cancellous-chondral bone across the lesion interface; stable union, defined as $\geq 50\%$ osseous continuity with partial fibrous tissue interposition but without signs of instability; or nonunion, defined as $< 50\%$ continuity with extensive fibrous tissue or persistent fluid signal beneath the lesion.³⁵ Less commonly, when advanced imaging (ie, MRI or CT) was not obtained at final follow-up, healing status was assessed using plain radiographs.

Additional outcomes included union status at early postoperative advanced imaging, need for reoperation for either planned (ie, routine hardware removal) or

unplanned events (ie, hardware failure), and secondary articular surface wear noted at the time of reoperation. Secondary cartilage surface was graded using the International Cartilage Repair Society (ICRS) classification system based on review of operative reports and arthroscopic images, and any changes greater than grade 1 were reported.

Statistical Analysis

Continuous variables were assessed for normality using the Shapiro-Wilk test, and nonparametric comparisons were performed using the Mann-Whitney *U* test when appropriate. Categorical variables were summarized using contingency tables, with overall group differences assessed using the chi-square test or the Fisher exact test depending on normality. Statistical significance was defined as $P < .05$. All analyses were performed using *R* statistical software (Version 4.5.1; R Foundation for Statistical Computing).

TABLE 1
Patient, Lesion, and Surgical Characteristics Between Screw and Suture Bridge Cohorts^a

MFC OCD Variables	Screws n = 29	Suture Bridge n = 40	P
Patient Characteristics			
Age, years	14.2 ± 2.7	14.4 ± 2.2	.723
Male	16 (55.2)	26 (65)	.409
BMI, kg/m ²	21.6 ± 5	24.4 ± 7.1	.159
Ethnicity			.501
White	20 (69)	25 (62.5)	
Black or African American	6 (20.7)	10 (25)	
Asian	1 (3.4)	0 (0)	
Multiracial	0 (0)	3 (7.5)	
Other	2 (6.9)	2 (5)	
Follow-up duration, months	22.1 ± 14.1	20.6 ± 11.7	.880
Preop OCD Characteristics			
Lesion surface area, cm ²	3.2 ± 1.4	4.4 ± 1.9	.029
Lesion depth, mm	8.7 ± 2	10 ± 1.6	.001
Varus alignment	3 (13.6)	8 (25)	.494
Presence of cyst >2 mm	10 (34.5)	10 (25)	.392
Bone within the progeny	27 (93.1)	39 (97.)	.568
Displaced progeny	21 (72.4)	21 (52.5)	.094
Oreo cookie	17 (60.7)	15 (45.7)	.236
Omen sign	20 (71.4)	27 (77.1)	.604
Surrounding edema, % of MFC			.002
No edema	0 (0)	2 (5)	
<25	5 (17.2)	7 (17.5)	
≥25-50	1 (3.5)	14 (35)	
≥50	23 (79.3)	17 (42.5)	
Surgical Factors			
Procedure details			.002
Drilling + fixation	18 (62.1)	9 (22.5)	
Drilling + debridement + fixation	0 (0)	2 (5)	
Drilling + debridement + bone graft + fixation	11 (37.9)	29 (72.5)	
Implant details			
Screws	29 (100)	0 (0)	
Absorbable suture	0 (0.0)	8 (20.0)	
Nonabsorbable suture	0 (0.0)	32 (80.)	
Surgical approach			.004
Arthroscopic	20 (69)	13 (32.)	
Open	9 (31.0)	27 (67.5)	
Number of implants across lesion ^b	2.9 ± 1.1	3.1 ± 1	.261
Use of BMAC	23 (79.3)	9 (22.5)	<.001
Concomitant procedures	0 (0.0)	8 (20)	.017

^aData are presented as mean ± SD or n (%). Boldface P values indicate statistical significance. BMI, body mass index; BMAC, Bone marrow aspirate concentrate; MFC, medial femoral condyle; OCD, osteochondritis dissecans; Preop, preoperative.

^bNo. of screws or suture limbs across lesion.

RESULTS

Cohort Characteristics and Imaging Availability

A total of 69 unstable MFC OCD lesions treated with internal fixation were identified in 64 patients with a minimum of 6-month follow-up. Overall, the mean age of the study population was 14.3 ± 2.4 years, 42 of 69 (60.9%) patients were male, and the mean follow-up duration was 21.3 months (range, 6-61.5 months). Baseline patient, lesion, and surgical characteristics are summarized for each group in Table 1.

Preoperative MRI was available for all 29 lesions treated with screw fixation and all 40 lesions treated with suture bridge fixation. Early postoperative advanced imaging was available for 56 of 69 lesions (81.2%) at a mean of 4.7 months after surgery (range, 2.3-8.4 months). In the screw fixation cohort, early postoperative imaging consisted of 12 MRI studies and 4 CT scans obtained at a mean of 4.9 ± 1.2 months after surgery. In the suture bridge cohort, early postoperative imaging included 38 MRI studies and 2 CT scans obtained at a mean of 4.6 ± 1.6 months after surgery. The difference in timing of early postoperative advanced imaging between groups was not

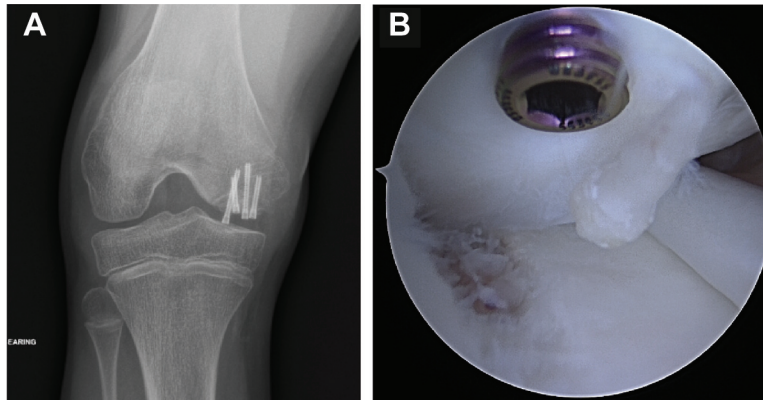


Figure 3. (A) Plain radiographic image demonstrating backout of a metallic headless compression screw in the application of MFC OCD fixation. (B) Arthroscopic image of the medial compartment demonstrating backout of the metallic headless compression screw at 6.5 months after initial fixation and associated ICRS grade 4 changes on the adjacent tibial articular surface. ICRS, International Cartilage Regeneration & Joint Preservation Society; MFC, medial femoral condyle; OCD, osteochondritis dissecans.

statistically significant ($P = .826$). An additional 24 advanced imaging studies were obtained at a mean of 20 months postoperatively (range, 6.1-61.5 months), including 10 lesions treated with screws (8 MRI, 2 CT) and 14 lesions treated with suture bridge fixation (all 14 MRI).

Patient, Lesion, and Surgical Characteristics

Patient characteristics were similar between lesions treated with screws and those treated with suture bridge fixation. Lesions treated with suture bridge fixation were significantly larger and deeper than those treated with screws (surface area: 4.4 ± 1.9 vs 3.2 ± 1.4 cm²; $P = .029$; depth: 10 ± 1.6 vs 8.7 ± 2 mm; $P = .001$), and had less surrounding bone marrow edema on preoperative MRI ($P = .002$). While not significantly different between groups, notable lesion characteristics included displaced progeny in 21 of 29 (72.4%) of screw fixation cases and 21 of 40 (52.5%) of suture bridge cases ($P = .094$), an “Oreo Cookie” sign in 17 of 29 (60.7%) and 15 of 40 (45.7%) of cases, respectively ($P = .236$), and an “Omen sign” in 20 of 29 (71.4%) of the screw fixation group and 27 of 40 (77.1%) of the suture bridge group ($P = .604$) (Table 1).

Suture bridge fixation was more frequently augmented with bone grafting (29/40 [72.5%] vs 11/29 [37.9%]; $P = .002$), whereas screw fixation more commonly involved drilling and use of Bone marrow aspirate concentrate (23/29 [79.23%] vs 9/40 [22.5%]; $P < .001$). Concomitant procedures were more frequent in the suture bridge cohort (8/40 [20%]), whereas none were performed in the screw cohort (0/29 [0%], $P = .017$). Among these procedures, 6 involved guided growth about the knee to modulate coronal plane alignment, 1 isolated medial patellofemoral ligament (MPFL) reconstruction, and 1 MPFL reconstruction combined with tibial tubercle osteotomy. In each case of patellar instability, the MFC OCD was judged to be independent of the patellofemoral pathology.

Implant Removal and Secondary Articular Surface Changes

Overall implant removal was more frequent after screw fixation than suture bridge fixation (26/29 [89.7%] vs 26/40 [65%]; $P = .024$). Within the suture bridge group, 8 of 40 (20%) of lesions were stabilized with absorbable suture and 32 of 40 (80%) of lesions with nonabsorbable suture. All nonabsorbable sutures were intended for later removal; however, a small number of families declined the procedure despite this recommendation. No lesions treated with absorbable suture underwent implant (ie, suture only) removal, while 26 of 32 (81.3%) of nonabsorbable suture constructs underwent suture removal. Accordingly, when screws were compared specifically to nonabsorbable suture constructs, the rate of permanent implant removal was not different (26/29 [89.7%] vs 26/32 [81.3%], respectively; $P = .478$). However, implant removal occurred earlier in the screw group (5.1 ± 4.1 vs 8.5 ± 7.8 months; $P = .019$).

Among lesions undergoing a second operation ($n = 28$ screws, 27 suture bridges), secondary articular surface assessment demonstrated significantly worse ICRS grades in the screw fixation group compared with the suture bridge group ($P < .001$). ICRS grade 4 changes were observed in 3 lesions (10.7%) after screw fixation (Figure 3). In contrast, only a single lesion in the suture bridge cohort demonstrated secondary surface wear, which was limited to ICRS grade 2. All observed secondary articular surface changes involved the adjacent medial tibial plateau.

Lesion Union and Conversion Rates

On early postoperative advanced imaging at a mean of 4.7 months after surgery, union status distribution differed between fixation techniques ($P = .014$). Of those with early postoperative advanced imaging, lesions rarely

TABLE 2
Outcomes After Screw Versus Suture Bridge Fixation for Unstable MFC OCD Lesions^a

MFC OCD Variables	Screws n = 29	Suture Bridge n = 40	P
Overall screw/suture removal ^b	26 (89.7)	26 (65)	.024
Screw/nonabsorbable suture removal	26 (89.7)	26 (81.3)	.478
Time to fixation removal, months	5.1 ± 4.1	8.5 ± 7.8	.019
ICRS grade of secondary articular surface at time of second surgery ^c	n = 28	n = 27	<.001
2	0 (0)	1 (3.7)	
3	0 (0)	0 (0)	
4	3 (10.7)	0 (0)	
OCD status at early post op MRI/CT	n = 17	n = 40	.014
Fully united	4 (23.5)	1 (2.5)	
≥50% union	9 (52.9)	23 (57.5)	
<50% union	3 (17.7)	16 (40)	
Conversion to secondary cartilage restoration procedure	1 (5.9)	0 (0)	
OCD status on final imaging			≥.999
Fully united	13 (44.8)	18 (45)	
≥50% union	11 (37.9)	14 (35)	
<50% union	0 (0)	0 (0)	
Conversion to secondary cartilage restoration procedure	5 (17.2)	8 (20)	≥.999
Time to conversion, months	12.7 ± 8.7	12.6 ± 9.2	.833

^aData are presented as mean ± SD or n (%). CT, computed tomography; ICRS, International Cartilage Repair Society; MFC, medial femoral condyle; MRI, magnetic resonance imaging; OCD, osteochondritis dissecans.

^bSuture removal was only performed for nonabsorbable suture bridge constructs.

^cOnly assessed for grade ≥2.

demonstrated complete union, but this was more common in lesions treated with screw fixation (4/17 [23.5%] vs 1/40 [3.7%]) whereas lesions treated with suture bridge fixation more commonly demonstrated stable (≥50%) union (23/40 [57.5%] vs 9/17 [52.9%]) at this early time point. One lesion treated with screw fixation required early conversion (defined as before early postoperative advanced imaging study) to a secondary cartilage restoration procedure (osteochondral allograft). In contrast, no lesions in the suture bridge cohort underwent early conversion to a secondary surgery (Table 2).

Rates of conversion to a secondary cartilage restoration procedure (osteochondral allograft was uniformly performed) were no different comparing screw and suture bridge fixation (5/29 [17.2%] vs 8/40 [20%], respectively; $P \geq .999$), and the timing of conversion did not differ between groups (12.7 ± 8.7 vs 12.6 ± 9.2 months, respectively; $P = .833$). There was no difference in the overall distribution of final union status between fixation groups ($P \geq .999$) (Table 2). Full osseous union was achieved in 13 of 29 (44.8%) screw-treated lesions and 18 of 40 (45.0%) suture bridge-treated lesions; stable ≥50% union in 11 of 29 (37.9%) and 14 of 40 (35%), respectively. No lesions resulted in <50% union on final imaging.

Patient-Reported Outcome Measures

PROM data were available for 45 Pediatric International Knee Documentation Committee (Pedi-IKDC) (19 [65.5%]

screws, 26 [65%] suture bridge), 47 Knee injury and Osteoarthritis Outcome Score (KOOS) (20 [69%] screws, 27 [67.5%] suture bridge), and 44 Pediatric Functional Activity Brief Scale (Pedi-FABS) assessments (21 [72.4%] screws, 23 [57.5%] suture bridge). There were no statistically significant differences between fixation groups in change from baseline in all PROMs, including Pedi-IKDC, KOOS subscores (Activities of Daily Living [ADL], Pain, Quality of Life [QoL], Sports/Recreation, and Symptoms), and Pedi-FABS (all $P > .05$) (Figure 4).

When assessed across the entire study population, significant improvements were observed from pre- to postoperative follow-up in Pedi-IKDC (60 vs 82.9; $P < .001$), KOOS ADL (79 vs 93.2; $P < .001$), KOOS Pain (74.3 vs 90.3; $P < .001$), KOOS QoL (45.4 vs 69.8; $P < .001$), KOOS Sports/Recreation (56.9 vs 79.5; $P < .001$), and KOOS Symptoms (69.2 vs 85.7; $P < .001$). KOOS Pain Scale scores decreased appropriately (27.5 vs 17.4; $P = .008$) (Figure 5). There was no significant change in Pedi-FABS scores from preoperative to final follow-up (14.6 vs 14.8; $P = .890$). There were no statistically significant differences in final postoperative scores between groups across any final PROM domains (all $P > .05$).

Using previously established Pedi-IKDC and Pedi-FABS minimal clinically important difference (MCID) thresholds,³⁰ a greater proportion of screw-treated patients achieved the Pedi-IKDC MCID than suture bridge-treated patients (17/19 [89.5%] vs 17/26 [65.4%]). However, this difference did not reach statistical significance ($P = .086$). Pedi-FABS MCID achievement was not significantly

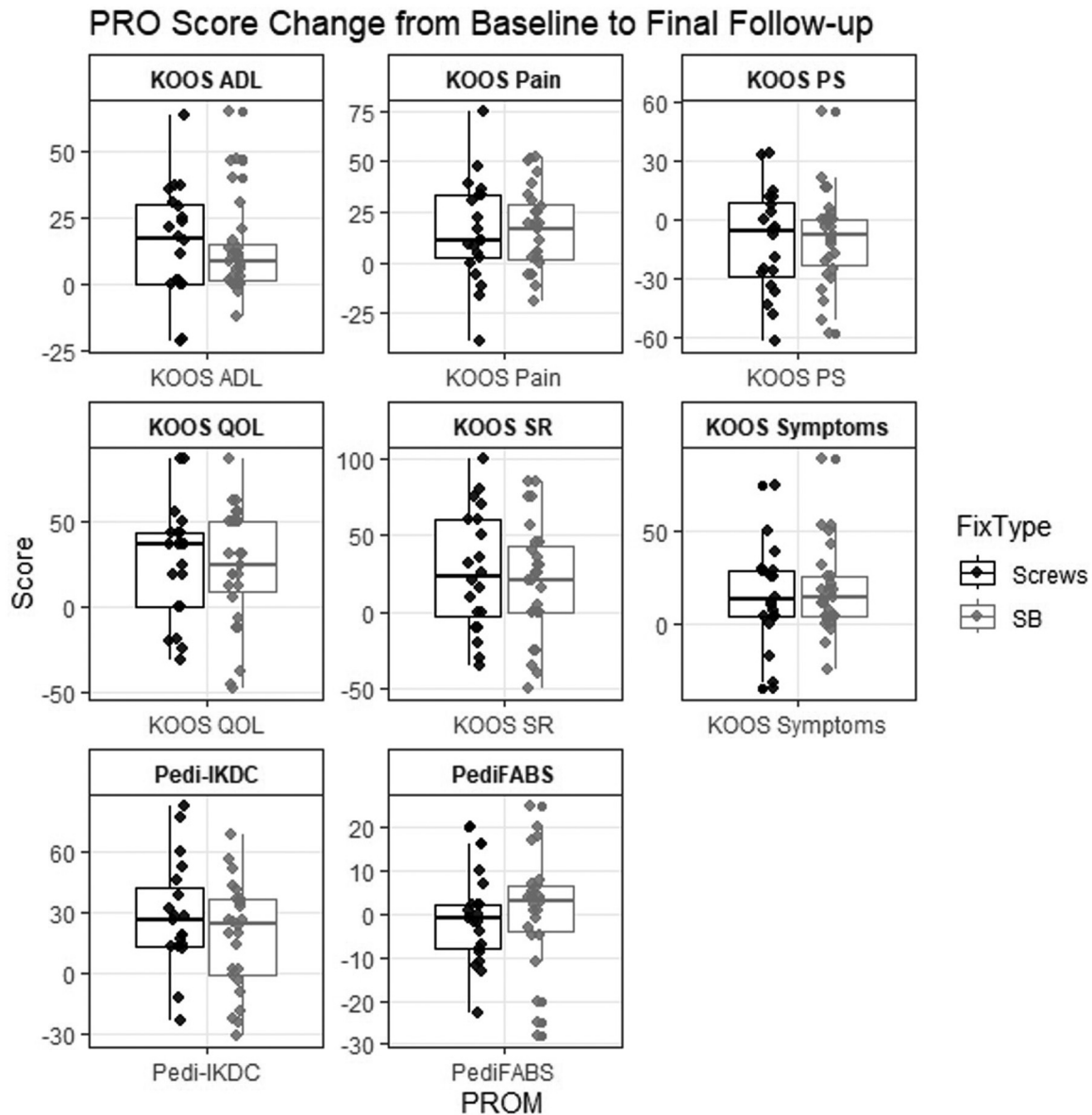


Figure 4. Patient-reported outcome score changes from baseline to final follow-up comparing screws versus SB. Box-and-whisker plots display median, interquartile range, and range, with individual patient data overlaid. No significant differences were observed between screws and SB for any outcome. Sample sizes differed by outcome measure: IKDC (screws = 19, SB = 26), KOOS (screws = 20, SB = 27), and Pedi-FABS (screws = 21, SB = 23). ADL, activities of daily living; KOOS-P, Knee injury and Osteoarthritis Outcome Score–Pain; Pedi-FABS, Pediatric Functional Activity Brief Scale; Pedi-IKDC, Pediatric International Knee Documentation Committee; PROM, patient-reported outcome measure; QOL, quality of life; SB, suture bridge; SR, sport and recreation.

different between groups (14/21 [66.7%] vs 19/23 [82.6%]; $P = .303$). KOOS MCID achievement²⁰ did not differ significantly between groups for any subscale (ADL: 60% vs 70.4%, $P = .541$; Pain: 40% vs 48.1%, $P = .767$; QOL: 60% vs 55.6%, $P \geq .999$; Sports/Recreation: 35% vs 25.9%, $P = .535$; Symptoms: 20% vs 22.2%; $P \geq .999$).

DISCUSSION

This study demonstrates similar rates of union and conversion to a secondary cartilage restoration procedure for unstable MFC OCD lesions between metallic headless compression screw and suture bridge constructs. However,

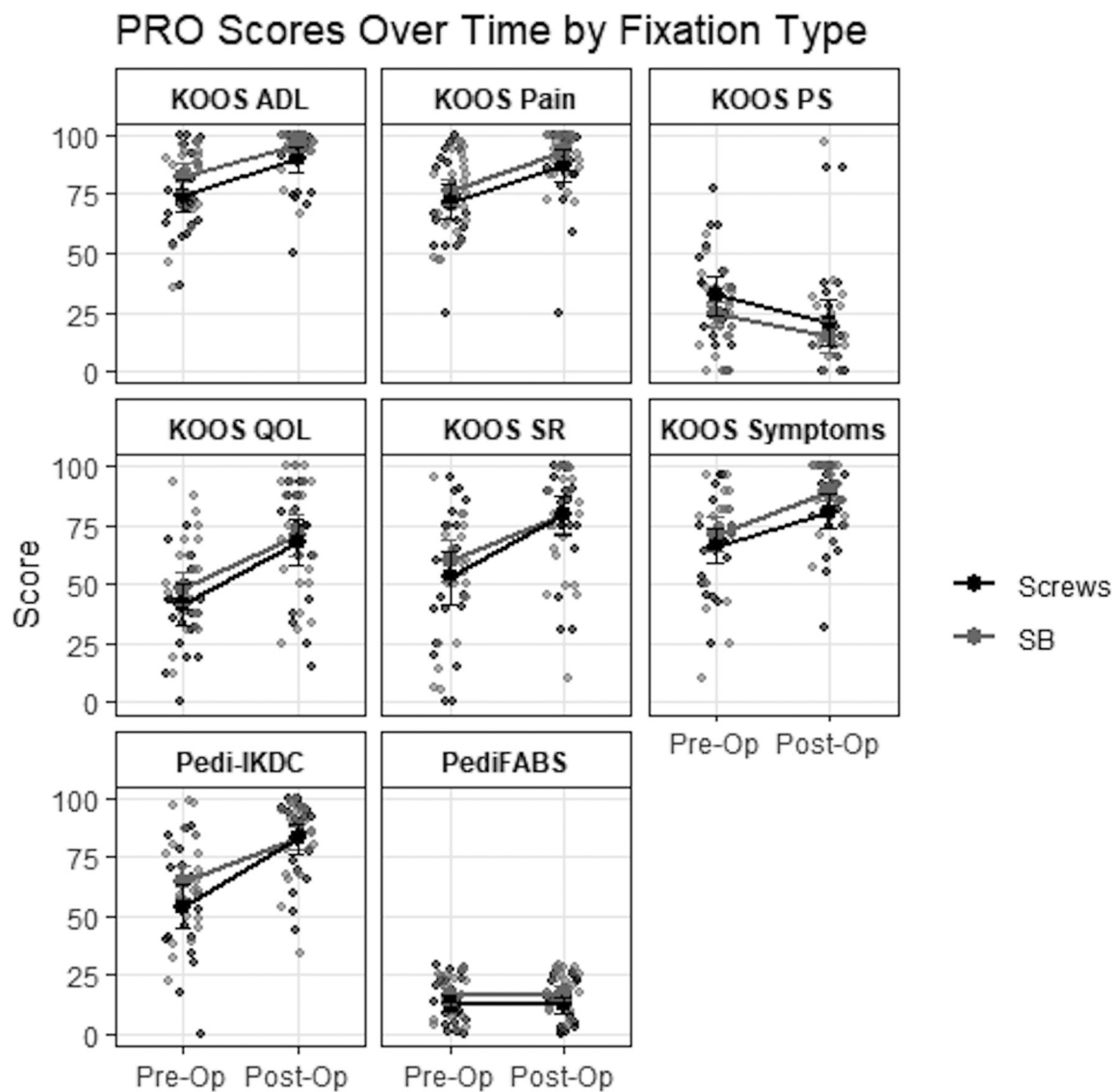


Figure 5. Patient-reported outcome scores at pre- to postoperative time points comparing screws and SB. Points represent individual patient data, with lines and markers indicating group means over time. Significant differences between pre- and postoperative time points were observed for all outcomes except Pedi-FABS. Sample sizes differed by outcome measure: IKDC (screws = 19, SB = 26), KOOS (screws = 20, SB = 27), and Pedi-FABS (screws = 21, SB = 23). ADL, activities of daily living; KOOS-P, Knee injury and Osteoarthritis Outcome Score–Pain; Pedi-FABS, Pediatric Functional Activity Brief Scale; Pedi-IKDC, Pediatric International Knee Documentation Committee; Postop, postoperative; Preop, preoperative; QoL, quality of life; SB, suture bridge; SR, sport and recreation.

screw fixation was associated with earlier implant removal and more severe secondary articular surface damage. There were no statistically significant differences in improvement from baseline to final PROMs between fixation groups.

The comparable healing outcomes with suture bridge constructs observed in this study substantiate the viability of suture bridge fixation to provide sufficient stability and durable fixation to accomplish union, even in significantly larger MFC OCD lesions. Although screw fixation may generate greater focal interfragmentary compressive

forces, suture bridge constructs are postulated to distribute fixation forces more broadly across the lesion surface, with particular effectiveness in stabilizing the peripheral margins.^{18,27,35} This difference may be advantageous in lesions with limited osseous material in the progeny or irregular geometry, including predominantly chondral lesions in which screw fixation lacks sufficient bone stock to achieve compression.

Screw fixation achieved a greater percentage of fully united lesions on early postoperative advanced imaging compared with the suture bridge group (4/17 [23.5%] vs

1/40 [2.5%]; $P = .014$). Despite this early finding, there was no difference in final union rates between the 2 fixation techniques. The perceived advantage of transarticular screw fixation may be outweighed by the potential risk of chondral injury, particularly with prolonged implant retention. Previous work has reported secondary cartilage wear, implant prominence, and catastrophic fixation failure associated with screw constructs, which can lead to unplanned surgery, including early implant removal.^{7,17,19,28,32} Consistent with these findings, screw fixation in the present study was associated with higher rates of implant removal and worse secondary articular surface changes, including ICRS grade 4 defects observed exclusively in the screw cohort (10.7%). This raises concern regarding the potential long-term chondral consequences of transarticular screw constructs.

In contrast, suture bridge fixation with nonabsorbable suture allowed for significantly longer implant retention compared with screw fixation (8.5 ± 7.8 vs 5.1 ± 4.1 months; $P = .019$). This prolonged retention was not associated with an increase in adverse articular surface changes. No second-look or suture removal procedures were performed for lesions treated with absorbable suture, as absorbable material has not been shown to generate chondral wear on arthroscopic follow-up.³⁵ Only 1 of 27 (3.7%) of secondary procedures in the suture bridge group demonstrated articular wear, which was limited to only ICRS grade 2, which was noted at 3.2 months postoperatively. This observation supports the notion of reduced concern for intra-articular complications with suture bridge constructs, allowing for prolonged lesion stabilization.³⁵ This may be advantageous in the management of biologically or mechanically unfavorable OCD lesions or in patients at risk for delayed or poor follow-up. In such circumstances, extended implant retention does not appear to pose a threat of higher-grade cartilage injury to the involved articular surfaces.

Advanced imaging played a central role in this study, although its use differed between fixation groups. CT imaging was favored over MRI in the screw fixation cohort secondary to metal artifact limiting MRI interpretability. In contrast, suture bridge constructs allowed routine postoperative MRI assessment without artifact, affording greater consistency in evaluating union by MRI, with CT reserved for supplemental evaluation of the lesion if desired. This difference represents a potential advantage of suture bridge fixation, as improved MRI utility may facilitate earlier identification of lesions demonstrating delayed or incomplete union.

Among lesions that ultimately converted to a secondary cartilage restoration procedure, the majority (7/13; 53.8%) demonstrated <50% healing on early postoperative advanced imaging obtained on average 4.7 months postoperatively. These findings are consistent with previous work demonstrating that the extent of early radiographic healing may be a useful predictor of failure after internal fixation of OCD lesions.^{1,13} Specifically, Hiramatsu et al¹³ demonstrated that lesions demonstrating less than approximately one-third radiographic healing or larger nonhealing areas on CT at 6 months postoperatively were at

significantly increased risk for reoperation. Notably, only 31.8% of lesions in their series were completely healed at 6 months, indicating that progression to union continues well beyond this early postoperative time period. Early postoperative advanced imaging may therefore serve as a useful adjunct for risk stratification, allowing identification of lesions unlikely to progress to union despite fixation.

The mean Pedi-IKDC and KOOS subscores improved significantly from pre- to postoperative assessment across all domains, with no statistically significant differences observed between fixation techniques. These improvements reached statistical significance, and most patients in both groups exceeded MCID thresholds for Pedi-IKDC and Pedi-FABS. These thresholds were established in a comparably aged pediatric anterior cruciate ligament and medial patellofemoral reconstruction cohort³⁰ and represent the closest available pediatric-specific approximation, as no OCD-specific MCID currently exists for these instruments. A greater proportion of screw-treated patients achieved the Pedi-IKDC MCID (89.5% vs 65.4%). Although this difference was not statistically significant ($P = .086$), it may reflect the modest sample size rather than a true absence of difference. No pediatric OCD-specific KOOS MCID exists; thresholds derived from a pediatric anterior cruciate ligament reconstruction cohort were therefore applied.²⁰ Despite the different indication, MCID achievement did not differ between groups under these thresholds. No statistically significant difference was observed in Pedi-FABS scores, suggesting that both techniques permit return to baseline activity levels. Collectively, these findings reinforce that while fixation strategy may influence surgical factors such as implant retention and articular surface preservation, both techniques are capable of achieving meaningful, patient-perceived improvement after treatment of unstable MFC OCD lesions.

Limitations

The cohort comparison design and evaluation of consecutive cases in this study introduce the potential for selection bias. Group heterogeneity represents an important limitation, as differences in surgical approach, biologic augmentation, and the addition of concomitant procedures reflect real-world, lesion-specific intraoperative decision-making. The increased use of open surgery in the suture bridge cohort is largely attributable to the technical requirements of the construct. Additionally, the contribution of individual surgical factors, including bone grafting and debridement, to union status cannot be fully isolated in this retrospective series, which is inherent to a consecutive clinical series. Concomitant guided growth procedures, performed in a subset of patients for coronal plane alignment correction as part of overall OCD management, represent an additional potential confounder that was not controlled for in this analysis.

The number of lesions with documented secondary surface changes was relatively small; however, the absence of

ICRS grade 4 changes in the suture bridge cohort compared with 3 cases in the screw cohort suggests a potential protective advantage of suture bridge fixation with respect to articular cartilage preservation, although larger studies are needed to confirm this finding. Additionally, the early postoperative advanced imaging modality differed between fixation groups, an inherent consequence of metal artifact limiting MRI interpretability in the screw fixation cohort, and may have influenced early healing assessments. Nevertheless, union was assessed using consistent criteria applied equivalently to both CT and MRI, and most screw fixation lesions underwent MRI after hardware removal as the final healing assessment. Given the rarity of this condition, this study represents a convenience sample, and our relatively small sample size means the analysis is likely underpowered; a lack of statistically significant difference should therefore not be interpreted as evidence that no true difference exists. Finally, PROMs were not available for all patients, which introduces the potential for response bias and may limit the interpretability and generalizability of the PROM findings. No pediatric OCD-specific MCID has been established for Pedi-IKDC, Pedi-FABS, or KOOS. The closest available pediatric-specific thresholds were applied for each of these variables,^{20,30} which may not fully generalize to this cohort.

CONCLUSION

Metallic headless compression screw and suture bridge fixation demonstrated comparable union rates, equivalent improvements in PROMs, and no statistically significant difference in conversion to a secondary cartilage restoration procedure in unstable MFC OCD lesions. Suture bridge fixation allowed longer implant retention with reduced secondary articular surface damage, making it a viable alternative to traditional screw fixation.

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